

A Report

On

Bank Portal Authentication

Submitted To

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In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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Batch:

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CANDIDATE'S DECLARATION

I, Rishikant Sharma, Roll No. CS-23411287, do hereby declare that the internship report titled “Bank Portal Authentication” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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I express my gratitude to my parents for their blessings.

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INTERNSHIP COMPLETION CERTIFICATE



Certificate of Completion – Cybersecurity Job Simulation, TATA (issued via Forage) to Rishikant Sharma, August 24, 2026

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4. PROJECT DESCRIPTION

4.1 Introduction

In today's digital banking ecosystem, security and user experience have become the two most crucial requirements for any financial institution. With the rise of cyber-attacks, phishing attempts, and unauthorized access, traditional username-password authentication is no longer sufficient to safeguard sensitive customer information. Modern banking systems must ensure that users can access their accounts securely, quickly, and conveniently, without compromising on safety.

To address these challenges, this project introduces a Secure Multi-Factor Authentication (MFA) system for a bank portal, named “SecureBank”, designed to provide robust security while maintaining a smooth and user-friendly experience. The system integrates multiple layers of authentication such as email OTP verification, QR-based device linking, and biometric authentication through Face ID. This ensures that even if one layer is compromised, unauthorized users still cannot access the banking account.

The solution begins with a simple registration process and email OTP verification to confirm the user's identity. Once the initial login is completed, the system generates a unique QR code that the user scans using a dedicated mobile application, securely linking the mobile device to the bank account. For seamless future access, the mobile app also guides the user through an Initial Face ID setup. On subsequent logins, the user enters credentials on the desktop, and a verification request is sent to the linked mobile device; access is granted only after the user successfully verifies their identity via Face ID. By combining email OTP, QR-based device pairing, and biometric verification, this project creates a highly secure yet convenient authentication framework tailored for modern banking needs.

4.2 Organization Profile

This internship project was carried out under the academic guidance of the School of Computer Science and Engineering, IILM University, Greater Noida, Uttar Pradesh, under the supervision of Mrs. Lalita Chaudhary. The work was undertaken as part of a structured internship program focused on building practical, industry-relevant skills in secure software development, with an emphasis on authentication systems used in the banking and financial-services domain.

(If this internship was hosted by an external company or organization, replace this section with that organization's name, its area of business, and a brief profile of the team or department you worked with.)

4.3 Problem Statement

Single-factor authentication (SFA), which relies exclusively on passwords, is fundamentally insecure in the modern threat landscape. Passwords can be stolen, guessed, or cracked through methods such as phishing, social engineering, keylogging, and brute-force attacks. The reliance on SFA for critical systems like bank portals creates an unacceptable level of risk, potentially leading to financial fraud, identity theft, and loss of customer trust. There is therefore an urgent need to develop and implement an advanced authentication system that is resilient to these threats while remaining accessible and convenient for diverse users.

4.4 Project Objectives

- To conduct a comprehensive review of existing authentication technologies and identify their strengths and weaknesses.
- To design the architecture for a secure Multi-Factor Authentication system incorporating knowledge, possession, and inherence factors.
- To develop a functional prototype of a bank portal login flow implementing password, OTP/QR-based device linking, and Face ID biometric verification.
- To evaluate the effectiveness of the implemented system against common attack vectors through simulated testing.
- To analyze the usability of the system to ensure a balance between enhanced security and user experience.

4.5 Scope of the Project

Scope:

- The project focuses on the authentication (login and registration) process of a web-based banking portal, “SecureBank”.
- The implementation includes three factors: password/OTP, QR-based mobile device linking, and Face ID biometric verification.

- The system was developed as a proof-of-concept prototype in a controlled environment.

Limitations:

- The project does not cover transaction authorization or other post-login security features.
- The biometric component relies on device/OS-level camera capture for Face ID rather than a custom-built facial recognition algorithm.
- Performance evaluation was conducted in a simulated environment rather than a live, large-scale production system.

4.6 Technologies and Tools Used

Software:

- Backend: Python (Django framework) for rapid, secure server-side development.
- Database: PostgreSQL for robust and secure data management.
- Frontend: JavaScript with the React.js library for a modern, responsive user interface.
- API Testing: Postman.
- Version Control: Git & GitHub.

Hardware:

- A development PC/laptop with a minimum of 8GB RAM.
- A smartphone with biometric (Face ID / fingerprint) capability for testing QR-linking and biometric login.

4.7 System Architecture

The proposed authentication workflow proceeds as follows: the user first registers on the portal and verifies their identity using a one-time password (OTP) sent to their email. On the next login, the system generates a unique QR code, which the user scans with the linked mobile application to pair the device to the account. The mobile app then guides the user through an initial Face ID setup. For all subsequent logins, the user enters credentials on the desktop, and the system sends a verification request to the

paired mobile device; the login completes only after the user authenticates using Face ID on that device.

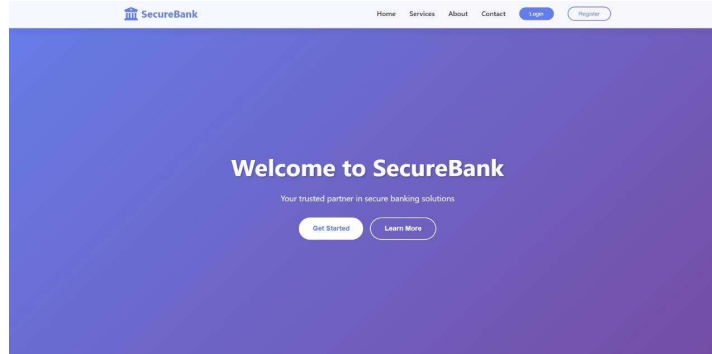


Fig 4.7.1: SecureBank portal landing page



Fig 4.7.2: Registration and login interface

4.8 Methodology

The project was developed using the Agile software development methodology, specifically the Scrum framework, broken down into two-week sprints, with each sprint focusing on a specific feature set (e.g., password module, OTP integration, QR-based device pairing, and biometric setup).

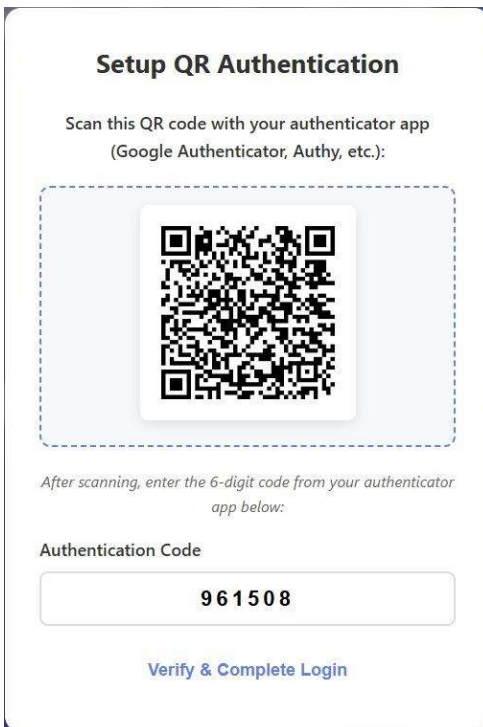
- Sprint 1 – Project Setup & User Model: Initialize the backend project and database schema for users, and implement secure password storage using hashing and salting.
- Sprint 2 – Basic Authentication & UI: Develop the registration and login pages, and build the username/password login API endpoint.
- Sprint 3 – OTP & QR Integration: Implement email OTP verification and generate a QR code for mobile device pairing.
- Sprint 4 – Biometric Integration: Implement Face ID capture and verification on the linked mobile device.

- Sprint 5 – Integration & Testing: Combine all authentication methods into a single workflow and perform functional and security testing.
- Sprint 6 – Finalization & Documentation: Refine the UI/UX, conduct final testing, and prepare the project report.



The image shows a login screen titled "Login to SecureBank" with a close button (X) in the top right corner. It features two input fields: "Email" and "Password". Below the password field is a "Login" button. At the bottom, there is a link that says "Don't have an account? Register here".

Fig 4.8.1: Email OTP verification screen



The image shows a screen titled "Setup QR Authentication". It instructs the user to "Scan this QR code with your authenticator app (Google Authenticator, Authy, etc.):". A QR code is displayed in the center. Below the QR code, it says "After scanning, enter the 6-digit code from your authenticator app below:". There is an input field for the "Authentication Code" which contains the text "961508". At the bottom, there is a button labeled "Verify & Complete Login".

Fig 4.8.2: QR-based device authentication setup

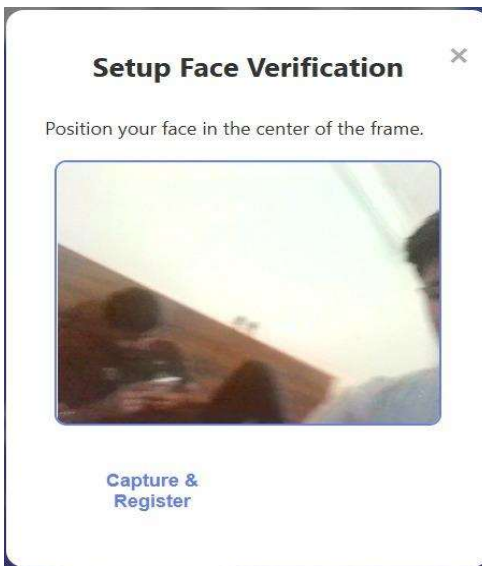


Fig 4.8.3: Face ID verification setup on the mobile application

4.9 Expected Outcomes

The primary outcome of the project is a fully functional prototype demonstrating the complete MFA login flow — password/OTP verification, QR-based device linking, and Face ID authentication. The security analysis is expected to show that the MFA system provides a quantifiable improvement over a traditional single-factor authentication (SFA) system, successfully blocking simulated unauthorized access attempts that would otherwise have succeeded against an SFA baseline. The usability analysis, based on login completion time and qualitative feedback, is expected to demonstrate that the added security layers do not impose an excessive burden on the user experience.

Overall, the project demonstrates that combining knowledge (password/OTP), possession (linked mobile device via QR), and inherence (Face ID) factors into a single, cohesive workflow significantly mitigates the risks associated with password-only authentication, while remaining convenient and intuitive for end users.

4.10 Certificates of Completion and Other Communication Mail Proof

[Attach the internship completion certificate issued by the supervisor/organization here, along with any relevant email communication proof.]

5. BIBLIOGRAPHY/REFERENCES

Books

1. Stallings, William. Cryptography and Network Security: Principles and Practice. Pearson, 7th Edition, 2017.

2. Kaufman, Charles, Radia Perlman, and Mike Speciner. Network Security: Private Communication in a Public World. Pearson, 2016.
3. Forouzan, Behrouz A. Data Communications and Networking. McGraw-Hill, 5th Edition, 2013.
4. Anderson, Ross. Security Engineering: A Guide to Building Dependable Distributed Systems. Wiley, 3rd Edition, 2020.

Research Papers

5. Aloul, Fadi, et al. "Two Factor Authentication Using Mobile Phones." IEEE/ACS International Conference on Computer Systems and Applications, 2009.
6. O'Gorman, Lawrence. "Comparing Passwords, Tokens, and Biometrics for User Authentication." Proceedings of the IEEE, 91(12), 2003.
7. Zhao, Wenyi, et al. "Face Recognition: A Literature Survey." ACM Computing Surveys, 35(4), 2003.
8. Jain, Anil K., Arun A. Ross, and Salil Prabhakar. "An Introduction to Biometric Recognition." IEEE Transactions on Circuits and Systems for Video Technology, 14(1), 2004.
9. Daugman, John. "How Iris Recognition Works." IEEE Transactions on Circuits and Systems for Video Technology, 14(1), 2004.

Web & Documentation References

10. OpenCV Documentation – <https://docs.opencv.org>
11. Flask Framework Documentation – <https://flask.palletsprojects.com>
12. Django Documentation – <https://docs.djangoproject.com>
13. TensorFlow Documentation – <https://www.tensorflow.org>
14. Scikit-Learn Documentation – <https://scikit-learn.org>
15. MySQL Documentation – <https://dev.mysql.com/doc>
16. SMTP/Email (Python smtplib) Documentation – <https://docs.python.org/3/library/smtplib.html>
17. OWASP Authentication Guidelines – <https://owasp.org/www-project-authentication-cheat-sheet>